Lean Canvas: AI-Powered Energy Optimization Agent

SDG 13 – Climate Action | IBM SkillsBuild AI/ML Internship – Final Project

1 PROBLEM

Top 3 problems

- **Wasted electricity:** idle appliances, poor scheduling and peak-hour usage drive up consumption in homes and buildings.
- **No visibility:** users see only a monthly bill, with no insight into when, where or why energy is wasted.
- **Climate impact:** higher consumption means higher CO_2 emissions, and users cannot measure or reduce their footprint easily.

Existing alternatives

Manual switch-off habits, basic smart plugs, utility bills and generic energy-saving tips.

4 SOLUTION

Top 3 features

- **ML demand forecasting** (LSTM / XGBoost) predicts hourly and daily usage.
- **Anomaly detection** (Isolation Forest / Autoencoder) flags abnormal spikes and faulty appliances.
- **Reinforcement-learning agent** (Q-learning / DQN) schedules and recommends actions to cut load while keeping comfort.

Plus: dashboard with a live CO_2-saved counter and plain-language tips from a GenAI layer (IBM watsonx.ai).

8 KEY METRICS

Key activities you measure

- % reduction in kWh consumed
- kg CO_2 avoided
- Forecast accuracy (MAE, RMSE, R^2)
- Anomalies detected and resolved
- Active users and retention
- Cost saved per user

3 UNIQUE VALUE PROPOSITION

Single, clear, compelling message

An AI agent that predicts, detects and cuts your energy waste automatically.

Lower bills and lower emissions, with no manual effort.

High-level concept

A smart-meter data pipeline with an ML "brain" that acts as a personal energy manager for every home, campus or office.

Why it is different

- Acts, not just reports
- Learns each building's own pattern
- Shows climate impact in real CO_2 numbers

5 UNFAIR ADVANTAGE

Can't be easily copied or bought

- Combined forecasting, anomaly detection and RL pipeline in one agent.
- Personalised models that improve as usage data grows.
- Explainable GenAI tips on IBM watsonx.ai / Granite models.

9 CHANNELS

Path to customers

- Web (Streamlit) and mobile dashboard
- Colleges and campus sustainability cells
- Smart-meter vendors and utility partners
- Social media and SDG awareness campaigns
- Open-source demo on GitHub

2 CUSTOMER SEGMENTS

Target customers

- Urban households with smart meters
- Colleges and university campuses
- Small offices and SMEs
- Facility and building managers

Early adopters

Eco-conscious households and campuses with sustainability targets and available meter data.

7 COST STRUCTURE

Customer acquisition costs, distribution, hosting, people

- **Hosting and compute:** cloud servers and model training / inference (IBM Cloud).
- **Data:** storage and smart-meter / IoT integration.
- **People:** ML engineering, development and maintenance.
- **Acquisition:** marketing, campus pilots and partnerships.

6 REVENUE STREAMS

Revenue model, lifetime value, revenue, gross margin

- **Freemium:** free basic dashboard, paid premium plan for advanced optimisation.
- **B2B licences:** per-building subscription for campuses and offices.
- **Partnerships:** utilities and smart-meter vendors (demand-response programmes).
- **Reporting:** CO_2 and sustainability reports for institutions.

Lifetime value comes from recurring subscriptions; software delivery keeps gross margin high at scale.

Numbers 1-9 follow the Lean Canvas filling order: Problem, Customer Segments, UVP, Solution, Unfair Advantage, Revenue, Cost, Key Metrics, Channels.